Master Study Notes: **Exploratory Data Analysis**

A Comprehensive Guide to Understanding, Profiling, and Visualizing Datasets

What is EDA?

Foundational Exploration. Essential for Model Success.

- Understand Data Structure & Types
- Detect Hidden Patterns & Insights
- Identify & Visualize Outliers & Anomalies
- Test Initial Hypotheses
- Check Critical Model Assumptions

> "**GARBAGE IN, GARBAGE OUT.** Skipping EDA is a Leading Cause of Model Failure in Production."

The Three Pillars of EDA

1. UNDERSTAND RELATIONSHIPS

Analyze how features connect to each other and to the target variable.

Practical Significance: Guides feature selection and identifies multi-collinearity, optimizing model inputs.

2. UNCOVER INNER DYNAMICS

Examine raw data for hidden patterns and structural anomalies.

Practical Significance: Informs data processing strategies, missing data handling, and reveals feature interactions.

3. SPOT OUTLIERS

Identify extreme values deviating from the main data cluster.

Practical Significance: Prevents algorithmic skew and improves model robustness by removing or correcting highly influential points.

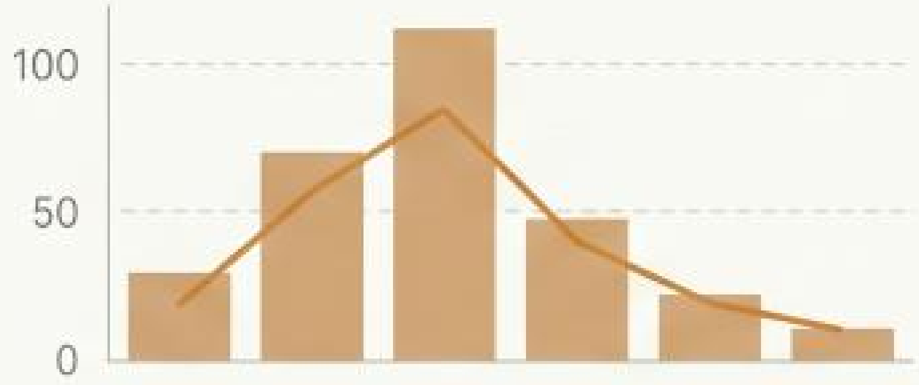
Data Categories: Know Your Variables

NUMERICAL DATA (Quantitative)

Mathematical operations apply

Examples

Age, Fare, Salary



Visualize with Histograms, Box plots, Scatter plots.
Calculate Mean, Median, and Standard Deviation.

CATEGORICAL DATA (Qualitative)

Distinct text labels or classes

Examples

Sex, Pclass, Embarked, Survived

Visualize with Countplots, Bar charts, Pie charts.
Calculate Frequencies and Mode.

TIME-SERIES DATA (Sequential)

Chronologically indexed points

Examples

Stock Prices, Daily Temperatures

Visualize with Line charts.
Analyze Trends, Seasonality, and Cyclical patterns.

Common Beginner Mistake: Misclassifying data types leads to applying incorrect tools and misleading insights.
Always verify your variable categories.

Univariate Analysis: Categorical Features

Examine one variable to describe its individual
Eɑamine one variable to describe its individual
behavior, determining counts and percentages of
each group.

Count & Frequency

```
counts = df['col'].value_counts()
```

Absolute Comparisons

```
df['col'].value_counts().plot(kind='bar', color='#D4A574')
```

Relative Percentages

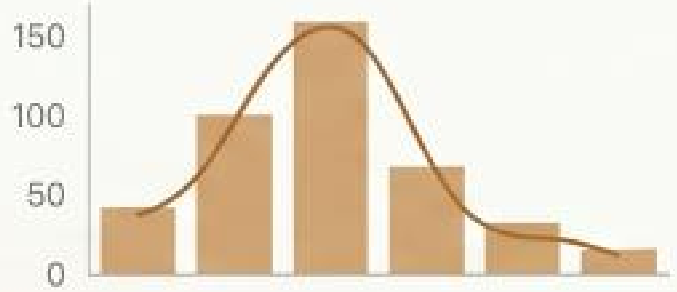
```
df['col'].value_counts(normalize=True).plot(kind='pie')
```

Full Distribution

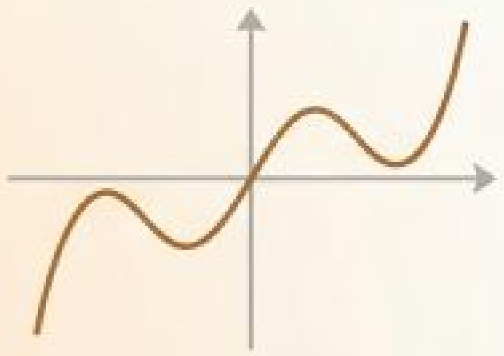
```
print(df['col'].value_counts(normalize=True))
```

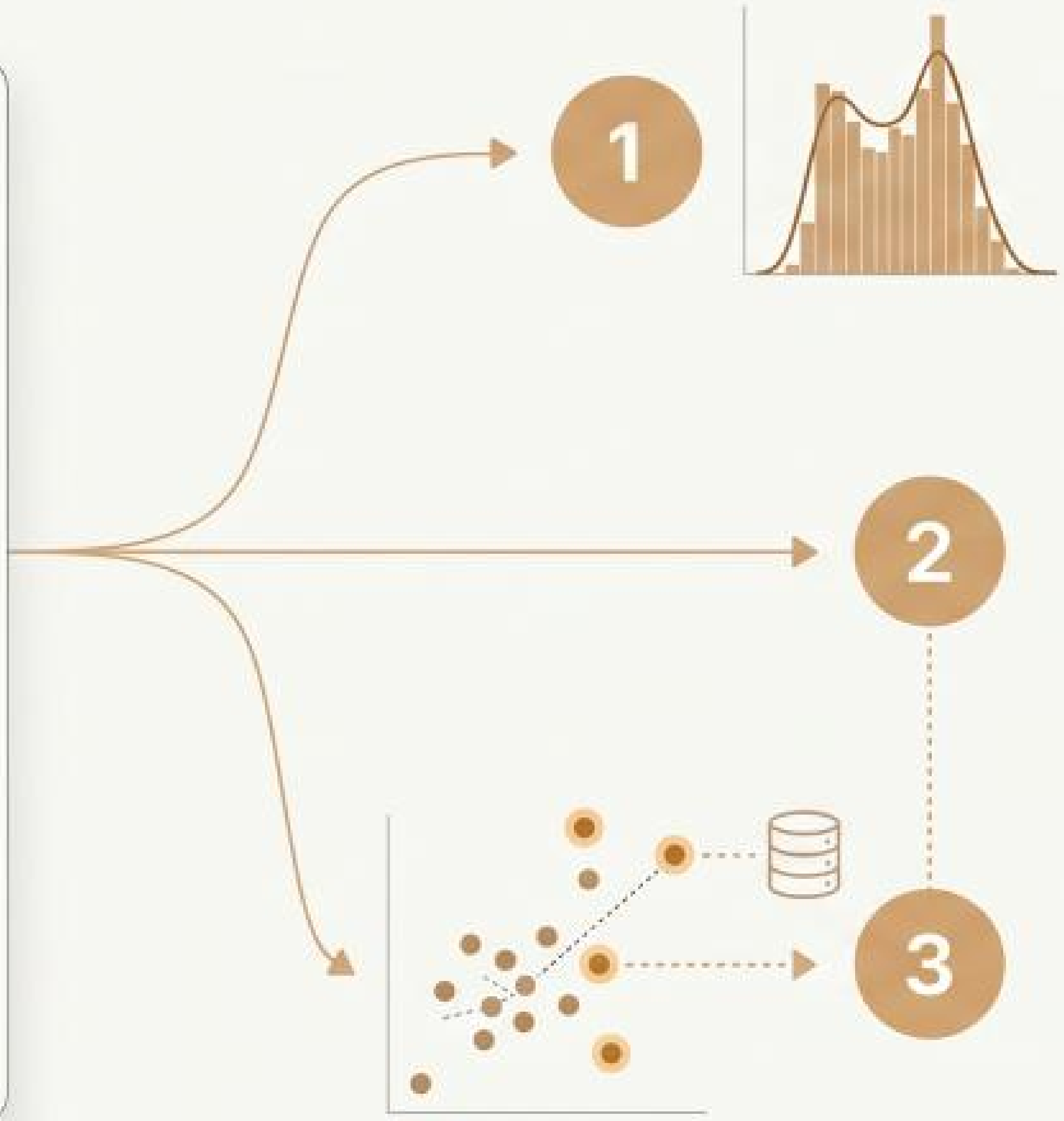
Univariate Analysis: Numerical Features

Analyzing single numerical variables: understanding distribution, tendencies, and spread.



Key Metrics:
.describe() provides count, mean, std, quartiles.
.skew() reveals symmetry.

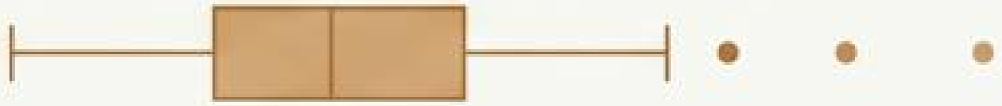




Distribution Shape

```
data['Feature'].plot(kind='hist', kde=True, bins=30)
sns.kdeplot(data['Feature'], color='#D4A574')
```

Central Tendency & Spread



```
data.boxplot(column=['Feature'], sym='#D4A574*')
```

Anomaly Extraction

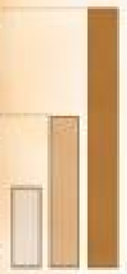
```
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1
Outliers = df[(df < (Q1 - 1.5 * IQR)) | (df > (Q3 + 1.5 * IQR))]
```

Bivariate Analysis: Categorical vs. Categorical Categorical

Exploring Relationships Between Categories

2. Normalized contingency (using normalize='columns')

	A	B	C
A	10	5	9
B	35	13%	16%
C	15%	15%	20%
D	15%	15%	25%



- Analyze distribution of variables within categories
- Use `**pd.crosstab(..., normalize='columns')**` to get column percentages.

```
normalized_crosstab = pd.crosstab(df['Category A'], df['Category B'], normalize='columns')
```

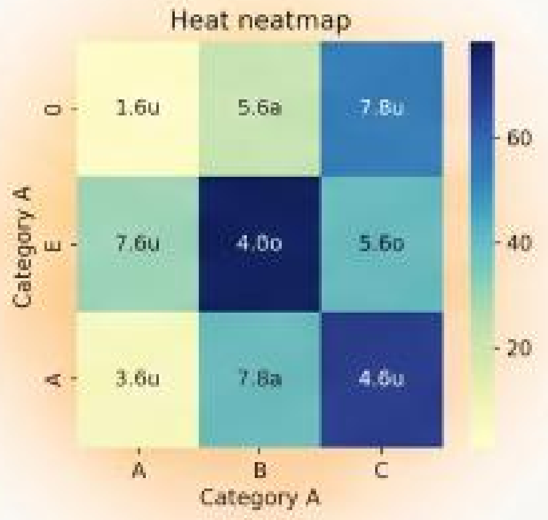
1. Cross-Tabulation (pd.crosstab)

	Category		
	A	B	C
A			
B			
C			

- Understand frequencies between two categorical variables
- Use `**pd.crosstab(df[var1], df[var2])**` for raw counts.

```
crosstab_df = pd.crosstab(df['Category A'], df['Category B'])
```

3. Optimal Visualization: Seaborn Heatmap



- Quickly identify high and low frequency pairs
- Color-coding for instant relationship analysis.

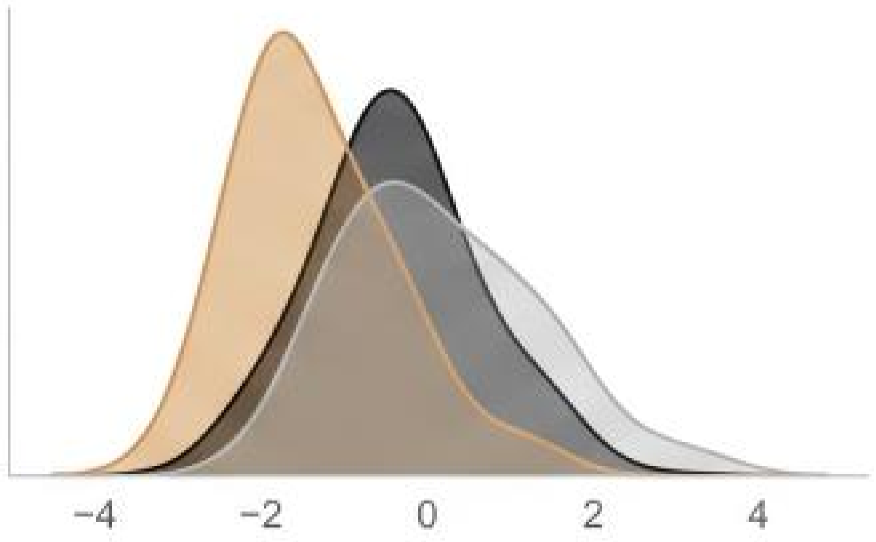
```
sns.heatmap(normalized_crosstab, annot=True, cmap='YlGnBu')
```

```
Sample production code:
df = annotated(heatmap((map),
      sns.crosstab(df['Category A', cmap=True')
sns.heatmap(nornalized_crosstab, annot=True,
cmap='YlGnBu')
```

Bivariate Analysis: Numerical vs. Categorical

Analyzing numerical metrics across categorical classes reveals group-level differences in probability distributions and central tendencies.

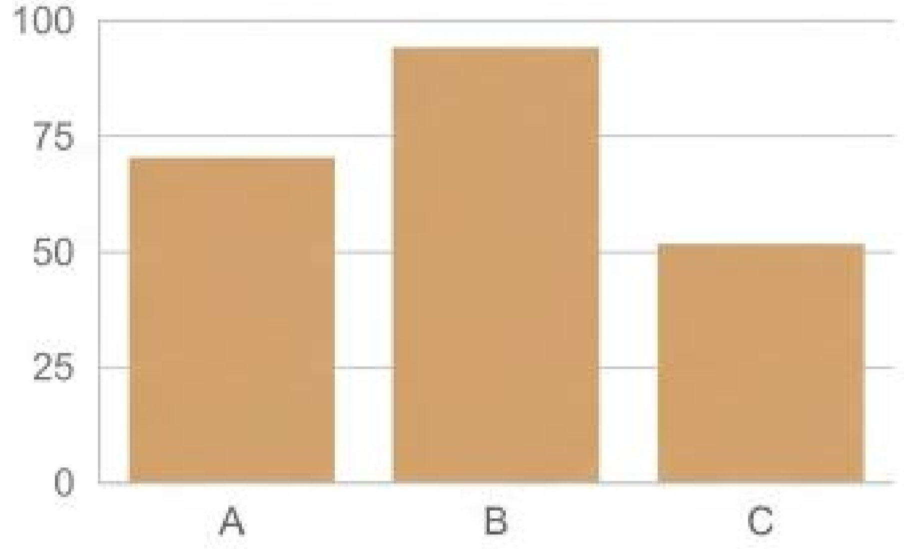
Overlaid KDE Plots



Compare entire probability curves.

```
sns.kdeplot(data=df, x="numeric_col",
hue="categorical_col", fill=True,
common_norm=False,
palette=["#D4A574", "#1D1D1F","#86868B"])
```

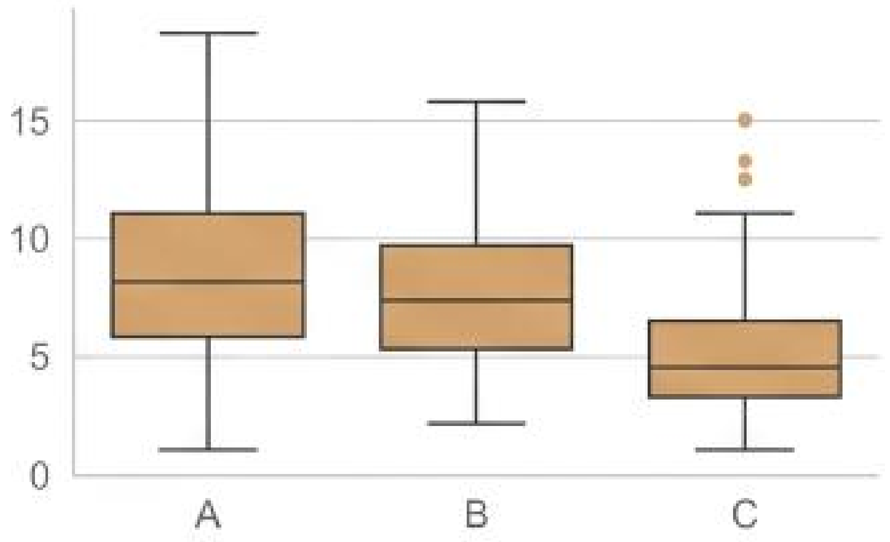
sns.barplot (Means)



Compare aggregate central tendencies (means).

```
sns.barplot(data=df, x="categorical_col",
y="numeric_col", estimator="mean",
color="#D4A574", errorbar=None)
```

sns.boxplot (Distributions)



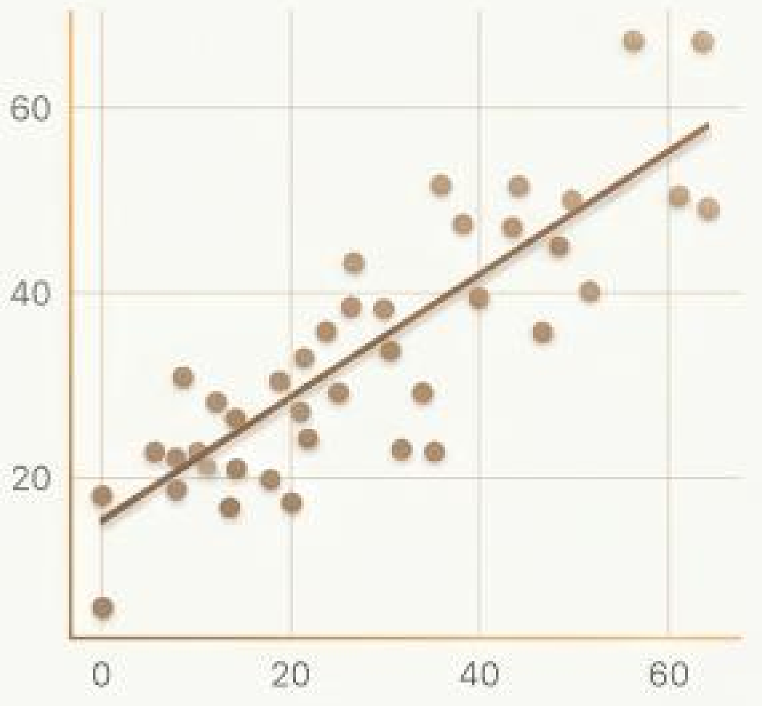
Compare statistical quartiles and outlier boundaries.

```
sns.boxplot(data=df, x="categorical_col",
y="numeric_col", color="#D4A574")
```

Bivariate & Multivariate: Numerical Relationships

Analyzing numerical relationships and identifying core **correlations**, trends, and trajectories.

Basic Scatter Plot



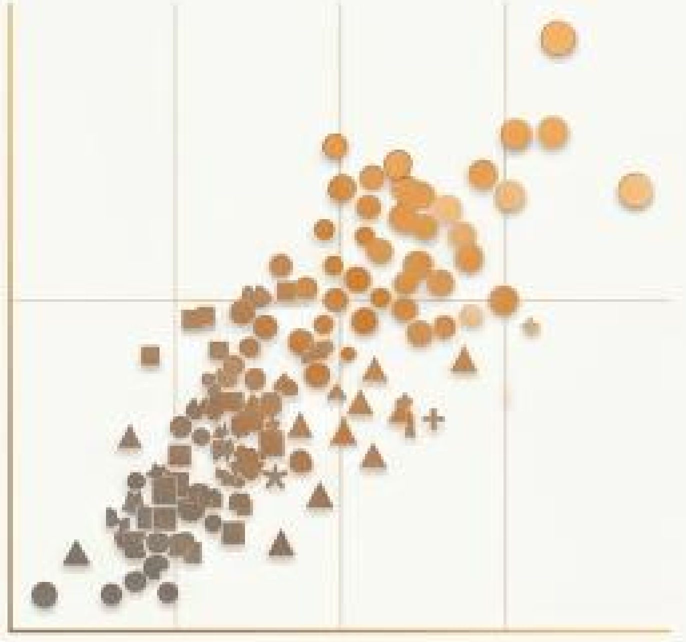

```
sns.scatterplot(x, y, data=df)
```

Maps individual data points on a 2D Cartesian plane to dina and intenttal plane to identify correlations.

- Linear indiuatc plots
- Point scatcies
- Linear trend ımt lines

Covering correlation analysis, scatter plots, and multivariate expansion using hue, style, and size.
Allows deeper high-dimensional insights.

Advanced Multivariate Scatter



```
sns.scatterplot(x, y, hue=h,
 style=s, size=v, data=df)
```

Extracts deeper insights from high-dimensional datasets without dimensiy reduction.

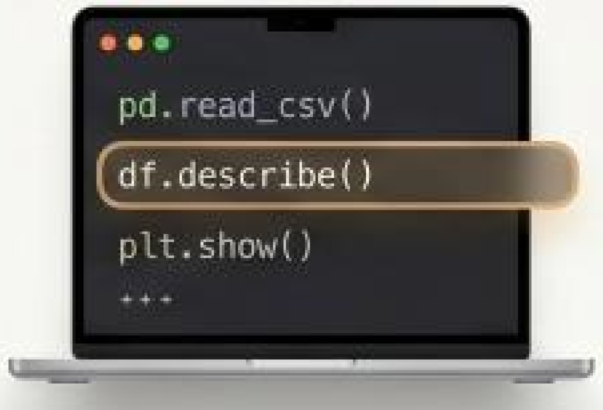
- Accent accent, gradient
- Point shape (different)
- Point size represent third

Covering correlation analysis, scatter plots, and multivariate expansion using hue, style, and size.
Allows **deeper high-dimensional insights**.

The Modern EDA Toolkit

Beyond Manual Coding: A balanced landscape of efficiency and deep insight.

- CORE LIBRARIES: Pandas, Matplotlib, Seaborn, Plotly.
- AUTOMATED EDA: Generate reports with functions like YData Profiling.
- COMPARATIVE APPROACH: Balancing full control with fast exploration.



Manual EDA

Manual Control for Deeper Domain Insights.

Automated EDA

Instant Reports with Statistics & Visualizations.

FAST START

Speed and Consistency

Consistency & Initial Data Quality Warnings.

SHAREABLE REPORTS

stakeholder-Friendly Insights & Collaboration.

The EDA Workflow in Practice

EDA IS ITERATIVE, NOT LINEAR

1. Dataset Overview

- Characterize Observations & Feature Types
- Identify Missing Rates & Duplicate Records

2. Feature Assessment

- Analyze Individual Feature Characteristics
- Descriptive Statistics & Distributions
- Evaluate Relevance

3. Data Quality Evaluation

- Spot Outliers & Inconsistencies
- Assess Skewness & Class Imbalance
- Map Missing Value Patterns

Best Practices

- Always start with overview statistics
- Visualize data before modeling
- Document all findings throughout
- Treat EDA as an ongoing process

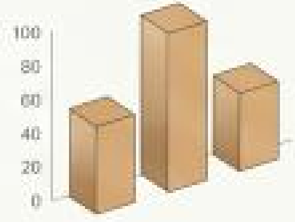
The Ultimate EDA Cheat Sheet

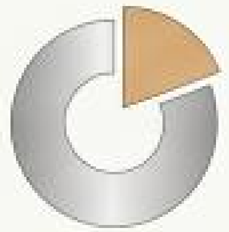
Variable Type Combination

Variable Count & Type

Univariate

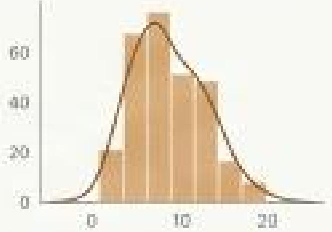
Categorical — Single Categorical

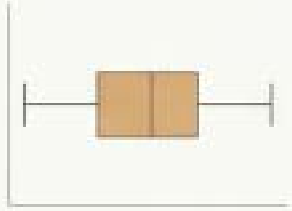




- value_counts
- bar charts
- pie charts

Numerical — Single Numerical

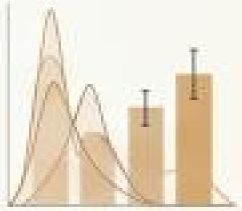


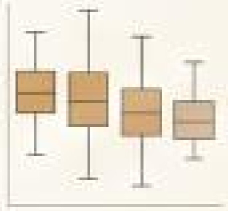


- describe, skew
- histograms
- KDE plots, boxplots

Mixed — Numerical vs. Categorical

- groupby mean
- overlaid KDE
- barplots
- boxplots





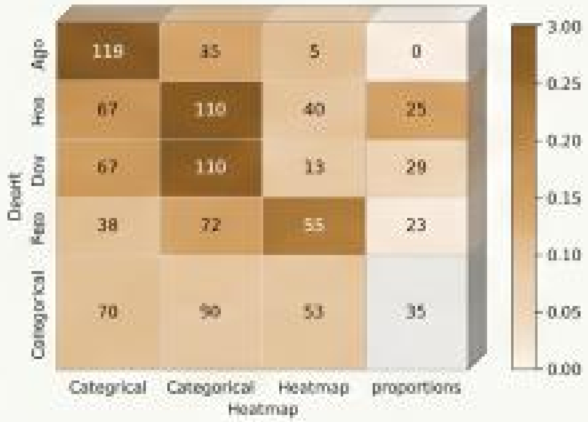
Mixed/Complex — Complex Data

- conditional mapping
- multivariate visualization

An integrated sensor for finding intricate patterns across diverse data types.

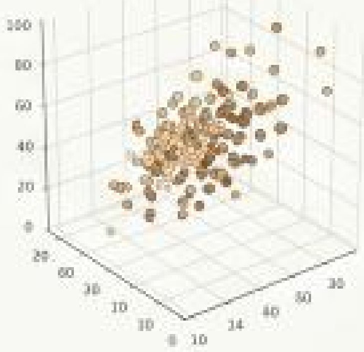
Bivariate / Multivariate

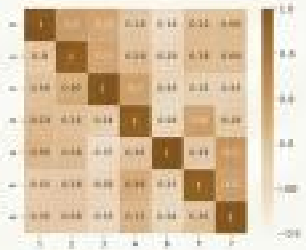
Categorical vs. Categorical



- pd.crosstab
- heatmaps

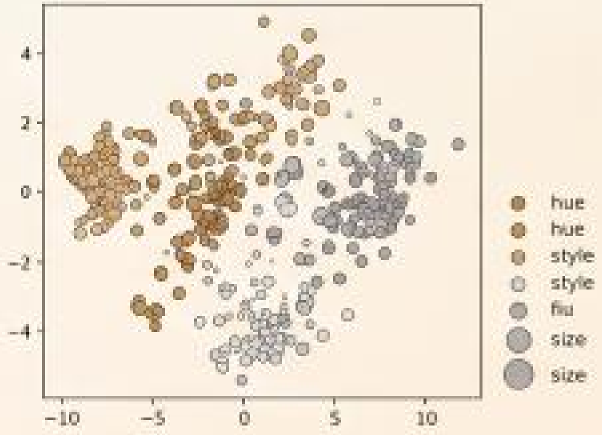
Numerical vs. Numerical





- corr
- scatter plots

Complex Data



- conditional mapping
- multivariate visualization